

A Spin-Dependent Fifth Force at 3.7 mm from a QCD-Anchored θ -Seesaw: Prediction, Bounds, and a No-Go Feasibility Study of the Shielded Nuclear-Spin Phase Detector

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<https://github.com/infosave2007/vmf>

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Abstract

The Null-Vector Gravity (NVG) framework, anchored to the single QCD input $M_{\Omega,0} = 859 \pm 8$ MeV, contains a pseudo-Goldstone mode θ of the vacuum condensate $\Phi = W e^{i\theta/f_a}$. In the θ -seesaw benchmark ($c_\nu = 1$) the decay constant $f_a = 1.07 \times 10^{11}$ GeV fixed by the neutrino sector yields a coherent pseudoscalar of mass $m_\theta = 53.2$ μ eV with Compton range $\lambda_\theta = 3.71$ mm and derivative (axial) couplings to nucleon spin, $g_{\theta pp} = -4.12 \times 10^{-12}$ and $g_{\theta nn} = -1.76 \times 10^{-13}$. We derive the resulting spin-dependent fifth force — a Moody–Wilczek tensor exchange, helicity suppressed by $(m_\theta/m_N)^2 \simeq 3.2 \times 10^{-27}$ relative to a scalar force of equal coupling — and show that the pion tail is dead at millimetre separations, leaving a clean window at $r \simeq \lambda_\theta$. Under the literature bounds quoted in Table 3 and the coupling conventions stated there, the benchmark lies below every bound, with margins that we quote separately (SN 1987A quadratic combination $\times 7.6 \times 10^4$; single nucleon couplings ~ 2 – 3 decades; white-dwarf electron channel $\times 5.8 \times 10^6$). A full finite-geometry treatment (Section 3.3) shows that for realistic bodies the force scales with the facing *area*, not with the product of bulk masses: the saturated force per unit area between fully polarized polyethylene slabs is $F/A \simeq 5.3 \times 10^{-24}$ N/m² at $d = \lambda_\theta$, rising by only $\times 2.2$ at a 371 μ m ($0.1\lambda_\theta$) gap. On this basis we propose a *shielded nuclear-spin phase detector*: the polarized composite source is separated by a superconducting membrane from a small hyperpolarized $^3\text{He}/^{129}\text{Xe}$ cell, and the exotic interaction is read as a pseudo-magnetic field — a shift of the spin precession phase or frequency — rather than as a mechanical torque. For the benchmark the pseudo-field at the cell is $B_{\text{exo}} \simeq 4.0 \times 10^{-29}$ T (^3He -equivalent), against a 30-day spin-projection floor of $\simeq 1.0 \times 10^{-24}$ T; the ^3He – ^{129}Xe species-ratio combination cancels real magnetic fields while the exotic shifts of the two species *add* (their effective charges carry opposite signs). The proposal removes the torsion fiber, the mechanical Q , the 0.1 Hz mechanical resonance, the large detector plate and the sub-millimetre mass drive; what remains — unavoidably — is the magnetic problem of the source: the polarized plates carry $\mu_0 M \simeq 1.4$ mT, and the unshielded fringe at the cell is $\simeq 1.4 \times 10^{-5}$ T even after isotope compensation. Closing on the benchmark requires a total suppression of 10^{21} – 10^{23} across the membrane factor S (ideal 2D ceiling $\simeq 10^8$ at this geometry), the in-situ servo and the comagnetometer rejection; the most optimistic ideal-component product ($\sim 10^{16}$) still misses by $\times 5 \times 10^4$ – 3×10^7 depending on the modulation mode, and no known physical mechanism closes the remaining gap. Together with the quantum gap of the phase readout itself (a factor $\times 2.6 \times 10^4$ to the spin-projection floor, i.e. $\times 7 \times 10^8$ in NT_2T) and the thermal gap of the torsion alternative, this constitutes a *no-go verdict* for the present source–detector family as a benchmark test (Section 5.7): the shielded nuclear-spin phase-readout architecture removes the mechanical limitations of torsion balances, but the benchmark remains inaccessible with the present source, spin-projection sensitivity, and demonstrated magnetic-suppression physics. The

instrument’s live content is metrology and exclusion: every *measured* suppression factor converts directly into exclusion reach ($\sim 3 \times 10^7 \times$ the benchmark coupling product at the optimistic component combination), and its staged commissioning — source magnetometry, shield factor $S(f)$, then the spin cell — is the content of the phased program of Section 5.6. A torsion-balance readout of the same source is retained as an alternative (Section 5.4): with the corrected force its benchmark torque is 1.2×10^{-21} N·m, a 30-day thermal SNR of 0.02 — an exclusion instrument, not a benchmark test. The program is locked to the companion haloscope effort — which thereby carries the benchmark-discovery burden — by a two-lab consistency condition: the 12.86 GHz microwave line and the 3.71 mm force range are the same mass m_θ , so a measurement in either laboratory fixes the other’s target with zero free parameters.

1 Introduction

The NVG/VMF framework [1] is anchored to one QCD input, $M_{\Omega,0} = 859 \pm 8$ MeV, and contains a pseudo-Goldstone mode θ of the vacuum condensate. In the θ -seesaw scenario [2], the same decay constant f_a that sets m_θ also generates Majorana neutrino masses through a Weinberg-like dimension-five operator suppressed by a two-loop QCD-anomaly coefficient; the benchmark $c_\nu = 1$ fixes $f_a = 1.07 \times 10^{11}$ GeV and yields the point target $m_\theta = 53.2$ μ eV. The companion experimental program is a microwave haloscope search for the coherent field oscillation at $f_\theta = m_\theta c^2/h = 12.86$ GHz.

The *same* couplings imply a second, independent laboratory signature: virtual θ exchange between spin-polarized bodies produces a spin-dependent fifth force of range $\lambda_\theta = \hbar/(m_\theta c) = 3.71$ mm. This paper (i) derives the force from the benchmark couplings without free parameters, (ii) audits its status against astrophysical bounds with honestly split margins, and (iii) audits the experimental feasibility of detecting it with the polarized composite source read by a *shielded nuclear-spin phase detector* — the exotic interaction read as a precession-phase shift of a small hyperpolarized nuclear-spin ensemble behind a superconducting membrane — or by a torsion balance. The verdict is a no-go for the benchmark with the present source–detector family, reached independently in the quantum, thermal and magnetic channels (Section 5.7); what the paper proposes is therefore the metrology-and-exclusion program of Section 5.6, quantified to the point where the conclusion is set by physics, not by engineering headroom.

Two negative statements frame the positive one. First, because the θ –nucleon coupling is axial-derivative, *unpolarized* matter feels no static force: there is no “mass-shielding” or anti-gravity channel in this framework, and earlier phenomenological estimates that relied on a free effective parameter η are superseded. Second, the real cosmological field cannot be resonantly detected by proton NMR at 12.86 GHz (which would require $B = 302$ T); the electron channel needs only 0.459 T but is loop-suppressed in the KSVZ-like benchmark. The virtual-exchange spin experiment audited here is therefore the candidate spin-side complement to the photon haloscope; the audit (Sections 5.2–5.7) concludes that it cannot cover the benchmark, leaving the haloscope carrying the discovery burden.

2 The θ -seesaw benchmark: derived couplings

With the single choice $c_\nu = 1$ the scenario has no further free parameters. Table 1 collects the quantities used below. The scenario band $c_\nu \in [1/3, 3]$ rescales them as $\lambda_\theta = 3.71 c_\nu$ mm and $g \propto 1/c_\nu$.

The product $m_\theta f_a$ agrees with the QCD axion band at 2.6%; we stress (as in [2]) that this is a normalization cross-check of the embedding, not an independent test.

Table 1: Derived θ -seesaw benchmark quantities ($c_\nu = 1$, KSVZ-like heavy-quark embedding).

Quantity	Value
Decay constant f_a	1.07×10^{11} GeV
Mass m_θ	53.2 μeV
Compton range λ_θ	3.71 mm
Oscillation frequency f_θ	12.86 GHz
Photon coupling $g_{\theta\gamma\gamma}$	2.08×10^{-14} GeV $^{-1}$
Isoscalar/isovector coefficients C_p, C_n	$-0.47, -0.02$
Proton coupling $g_{\theta pp} = C_p m_p / f_a$	-4.12×10^{-12}
Neutron coupling $g_{\theta nn} = C_n m_n / f_a$	-1.76×10^{-13}

3 The predicted force

3.1 Spin structure and suppression

A pseudoscalar with derivative coupling has *no* scalar Yukawa term $g^2 e^{-r/\lambda} / (4\pi r)$. The leading non-relativistic channel is the Moody–Wilczek tensor potential [3]: for two aligned spins $\vec{s}_1, \vec{s}_2$ separated by $\vec{r}$,

$$V_\theta(\vec{r}) = \frac{g_1 g_2}{16\pi m_1 m_2} \left[(\vec{s}_1 \cdot \vec{s}_2) \left(\frac{1}{r\lambda} + \frac{1}{r^2} \right) + (\vec{s}_1 \cdot \hat{r}) (\vec{s}_2 \cdot \hat{r}) \left(\frac{1}{\lambda^2} + \frac{3}{r\lambda} + \frac{3}{r^2} \right) \right] e^{-r/\lambda}, \quad (1)$$

suppressed by two powers of the nucleon mass relative to a scalar force of equal coupling. Numerically the suppression factor is $(m_\theta/m_p)^2 = 3.2 \times 10^{-27}$, so the formal scalar comparison $\alpha_{\text{coup}} \sim 2.3 \times 10^{14}$ (a ratio of couplings $g^2/(4\pi G m_p^2)$) is *not* a force ratio: at $r = \lambda_\theta$ the potential energy of one aligned proton pair is $|V_\theta| \simeq 3.4 \times 10^{-75}$ J (2.1×10^{-50} μeV), i.e. $|V_\theta|/V_{\text{grav}} \simeq 6.8 \times 10^{-14}$ per pair — thirteen orders of magnitude *below* gravity at the nucleon level. The macroscopic force survives only through Avogadro-scale spin counts and the short range.

3.2 Macroscopic force and scaling

Summing (1) over fully polarized bodies in the pointlike approximation at $r = \lambda_\theta$ (audit script `generator_project/calc_spin_fifth_force.py`) gives the base case

$$F(10 \text{ g} + 10 \text{ g}, \text{pol} = 1, r = \lambda_\theta) = 9.84 \times 10^{-23} \text{ N}, \quad \tau(1 \text{ cm arm}) = 9.84 \times 10^{-25} \text{ N} \cdot \text{m}. \quad (2)$$

The scaling law *within the pointlike approximation* is

$$F \propto M_{\text{src}} M_{\text{det}} \text{pol}_{\text{src}} \text{pol}_{\text{det}} e^{-r/\lambda_\theta}. \quad (3)$$

This law is used only as a normalization anchor: for bodies whose linear size exceeds a few λ_θ it overestimates the force by orders of magnitude, because only spins within ~ 1 – 3 Compton ranges of the facing surfaces contribute. The corrected scaling is derived in Section 3.3. Two structural points: (a) the pion-exchange tail that dominates ordinary nuclear spin forces is suppressed by $e^{-m_\pi r}$ with $m_\pi r / \hbar c \simeq 2.6 \times 10^{12}$ at $r = \lambda_\theta$, so the millimetre window is clean; (b) because the coupling is to *nucleon* spin (the electron coefficient C_e is loop-suppressed in the KSVZ-like embedding), ferromagnetic electron polarization does not contribute — the enabling technology must be cryogenic *nuclear* polarization.

3.3 Finite-geometry correction

For bodies larger than λ_θ the force must be computed as the overlap convolution of (1) over the two volumes (audit script `generator_project/calc_slab_geometry.py`, calibrated to the

audited pair force at $r = \lambda_\theta$). With both slabs polarized along $\hat{\mathbf{z}}$ and lateral extent $\gg \lambda_\theta$ the planar problem reduces exactly to a one-dimensional kernel:

$$\frac{F}{A} = n_1 n_2 P^2 \int_0^{t_1+t_2} M(w) f_s(d+w) dw, \quad M(w) = \min(t_1, w) - \max(0, w - t_2), \quad (4)$$

where d is the gap, $t_{1,2}$ the slab thicknesses, $M(w)$ the triangle measure of source–detector depth pairs with $s_1 + s_2 = w$ (zero at the facing surfaces), and $f_s(u) = -dW/du$ the sheet force from the 2D-integrated Moody–Wilczek kernel $W(u) = \int d^2\boldsymbol{\rho} V_{\text{MW}}(\boldsymbol{\rho}, u)$. Because the kernel is only logarithmically divergent at small separation, there is no large near-field lever (an earlier midpoint-grid version of this convolution overweighted the facing-surface pairs and produced a spurious one; the kernel evaluation is validated against a direct 3D pair sum at grid precision). We take a concrete material benchmark: polyethylene, whose signal is carried entirely by proton spins ($n_p = 8.1 \times 10^{28} \text{ m}^{-3}$, $\mu_p = 2.793 \mu_N$, the latter *strengthening* the polarization of Table 4 at fixed B, T). The results:

- **Thickness saturation:** at gap $d = \lambda_\theta$, F/A reaches 85% of its asymptote at $t = 3\lambda_\theta$ and saturates by $t \sim 5\lambda_\theta$; bulk deeper than that is inert. The saturated force per unit facing area is

$$F/A \simeq 5.3 \times 10^{-24} \text{ N/m}^2 \quad (d = \lambda_\theta, \text{ pol} = 1), \quad (5)$$

with a model uncertainty of order $\times 10$ from the tensor angular structure.

- **Mass-scaling violation:** two realistic 10 g polyethylene cubes at $d = \lambda_\theta$ produce $2.6 \times 10^{-27} \text{ N}$, while the pointlike law quotes $9.8 \times 10^{-23} \text{ N}$: the point approximation *overestimates by* $\sim 4 \times 10^4$. The correct scaling is $F \propto A \text{ pol}^2$ (facing area), not $M_{\text{src}} M_{\text{det}}$; the active mass is $\sim 1.5 \text{ kg/m}^2$ of facing skin.
- **Gap lever:** at $d = 0.1 \lambda_\theta = 371 \text{ } \mu\text{m}$ the kernel gives $F/A \simeq 1.15 \times 10^{-23} \text{ N/m}^2$, an enhancement of only $\times 2.2$ over the λ_θ gap — the energy kernel is merely log-divergent, so sub- λ_θ gaps buy little signal at a steep price in gap control.

Structural note: throughout, both slabs are polarized along the plate normal ($\hat{\mathbf{z}}$), which maximizes the tensor $(\mathbf{s}_i \cdot \hat{\mathbf{r}})^2$ term for facing pairs; an in-plane polarization retains only the $(\mathbf{s}_1 \cdot \mathbf{s}_2)$ term and reduces the signal by a factor of order a few, while strongly suppressing the classical magnetic background of Section 3.4. The kernel identity $F/A = n_2[U(d) - U(d + t_2)]$ with $U(z) = n_1 \int_z^{z+t_1} W du$ is verified against a direct three-dimensional pair sum (0.3–1.6% at $d = \lambda_\theta$, where the 3D grid converges); the quoted $\times 10$ model uncertainty is the angular/tensor-structure systematic, not discretization. All torque and pseudo-field estimates below use these geometry-corrected values.

3.4 The classical magnetic background: compensation audit

Fully polarized polyethylene is not magnetically inert: it carries a real magnetization $M = n_p \mu_p P \simeq 1.1 \times 10^3 \text{ A/m}$ ($\mu_0 M \simeq 1.4 \text{ mT}$). Because the exotic and magnetic source densities are pointwise proportional (both $\propto n_p P$) and both interactions share the dipole–dipole tensor angular structure, no spatial patterning can separate them: only the range differs (Yukawa e^{-d/λ_θ} versus long-range), so only range-shape discrimination over a gap scan can work. The dedicated magnetostatics audit (`generator_project/calc_magnetic_compensation.py`, exact closed-form field of charged rectangular faces, quadrature over the detector faces, validated against the infinite-sheet limit) gives:

- **Exact cancellation in the ideal limit:** outside a uniformly magnetized slab $B = 0$, so the net force between two infinite uniformly magnetized slabs is identically zero; the often-quoted pressure $\mu_0 M^2/2 \simeq 0.8 \text{ Pa}$ is an internal stress between the facing charge faces that self-cancels across the four faces of the slabs. Every residual torque is therefore a finite-size effect.

- **Fringing dominates the torsion channel:** for 0.5 m source and 0.32 m detector plates, the lateral (torsion-channel) magnetic force under a ± 0.5 mm source translation is $\simeq 8.5 \times 10^{-7}$ N, essentially *flat* across the gap scan $d \in [0.1\lambda_\theta, \lambda_\theta]$. The modulated magnetic torque is $\tau_{\text{mag}} \simeq 8.5 \times 10^{-4}$ N·m (with the resonant gain) — $\sim 10^{18}$ times the exotic signal; its variation over the gap scan is only 4.5×10^{-6} N·m ($\sim 7 \times 10^{15}$ times $\Delta\tau_{\text{exo}}$). Its flatness in d must not be over-read: extracting the benchmark amplitude from a shape fit of $F(d) = A_{\text{exo}} \text{shape}(d/\lambda_\theta) + F_{\text{fringe}}(d)$ would require modelling the fringe — geometry, $M(\mathbf{r})$ inhomogeneity, edge defects, motion trajectory — to a relative precision of order 10^{-15} – 10^{-16} ; shape fitting in the present form constrains interactions many orders of magnitude stronger than the benchmark, it does not discover the benchmark.
- **The anti-parallel sandwich fails:** a source stack of two oppositely polarized plates (net dipole zero) leaves a *larger* residual, $\tau_{\text{mag}} \sim 4 \times 10^{-5}$ N·m, and requires a plate-to-plate balancing tolerance $|\delta| \lesssim 10^{-16}$ in magnetization $\times$ volume — not physically realizable.

Status, honestly: no tested geometry reduces the magnetic torque to the 10^{-19} N·m level, and the built-in depolarized-source null cannot separate exotic from magnetic: both amplitudes scale as $P_{\text{src}}P_{\text{det}}$ and both vanish together at zero polarization, so the null test verifies spin dependence but not its origin; only the gap-dependence shape discriminates. The fringe is smooth and computable in closed form; after a fringe-model fit constrained by gap and position metrology, the residual is limited by polarimetry: polarization knowledge at the 10^{-3} level leaves a residual $\sim 10^{-3} \times 4.5 \times 10^{-6} \simeq 5 \times 10^{-9}$ N·m. This is the realistic *magnetic-calibrated floor*: the torsion apparatus is a null instrument with reach $\sim 4 \times 10^{12} \times$ the benchmark coupling product in the maximum-reach geometry — new tabletop territory for millimetre-range spin-spin forces (competitiveness against the compilation of [7] is not claimed here), but not a benchmark test. Reaching the benchmark itself requires reducing the present magnetic-calibrated residual by a further $\sim 4 \times 10^{12}$ ($5 \times 10^{-9}/1.15 \times 10^{-21}$), or $\sim 5 \times 10^{10}$ to reach the aspirational 10^{-19} N·m level — equivalently, demonstrating an independently validated background-rejection observable of that power, i.e., a source with polarized nucleon spin density and vanishing net magnetization. The naive candidate — a nuclear antiferromagnet — fails on coarse graining: a boson of range $\lambda_\theta = 3.71$ mm averages over $\sim 10^7$ lattice constants and reads only the coarse-grained spin density, which atomic-scale compensation zeroes together with the magnetization, so the exotic source dies with the background (multipole suppression $\sim (a/\lambda_\theta)^2$). The proportionality must instead be broken in *composition* space, $\sum_i n_i \mu_i P_i \simeq 0$ while $\sum_i n_i g_i P_i \neq 0$; two computed candidate routes are presented in Section 3.5. Historical spin-pendulum work used spin/orbital-compensated materials [5, 7]; here the nuclear magnetization *is* the signal carrier, which is precisely why the problem is hard.

3.5 Breaking the proportionality: two computed candidate routes

The condition above — polarized nucleon spin density at vanishing net magnetization — is satisfiable because the magnetic moment of a nucleus and its pseudoscalar charge are different linear functionals of its spin structure: per fully polarized nucleus the effective charge is $g_A = g_{\theta pp} \sigma_p + g_{\theta nn} \sigma_n$, where $\sigma_{p,n} = \langle \Sigma_z \rangle_{p,n}$ are the proton and neutron spin fractions (the orbital part $\langle L \rangle$ does not couple to the spin-spin potential). Nuclei with negative moments (^{15}N , ^{29}Si , ^{129}Xe , ^3He), polarized with spin along the holding field, carry their magnetic moment *against* it, so their magnetization cancels the proton magnetization. Because $|g_{\theta pp}/g_{\theta nn}| = 23$, the effective charges are nuclear-structure sensitive — for ^3He the -5.6% proton spin content overcompensates the 87% neutron content and flips the sign of g_A relative to the naive $\sigma_n g_{\theta nn}$ estimate — but the retained exotic charge of the composite is set by the proton layer alone: the isotope-layer term is a percent-level correction whose sign is nucleus-dependent (Table 2). The isotope layer is therefore a magnetic canceller, not an exotic-charge booster. The

compensation acts between two macroscopic uniform densities, which the 3.71 mm field resolves fully — there is no multipole suppression of the kind that kills the antiferromagnet. All numbers in this subsection are from `generator_project/calc_spin_compensation.py` and `generator_project/calc_shield_attenuation.py`.

Table 2: Nuclear spin decomposition of the negative-moment isotope candidates: spin fractions $\sigma_{p,n} = \langle \Sigma_z \rangle_{p,n}$ at full polarization, measured moment μ_A , signed effective charge $g_A = g_{\theta pp}\sigma_p + g_{\theta nn}\sigma_n$, isotope number density in the carrier solid and realistic polarization. Sources of σ : ^{15}N Schmidt $p_{1/2}$ neutron hole (core-polarization uncertainty $\sim 50\%$); ^{29}Si and ^{129}Xe single-particle $s_{1/2}$ estimate $\sigma_n = 2\mu_A/g_n$ (shell-model quenching lowers ^{129}Xe to ~ 0.33); ^3He S/D-state decomposition [9] ($\langle L \rangle \simeq 0.19$). “Effect” is the sign of the isotope layer’s exotic force on a proton detector relative to the proton layer.

Isotope	σ_p	σ_n	μ_A/μ_N	g_A	effect	$n_A, \text{m}^{-3} (P_A)$
^{15}N (Na^{15}N_3)	0	$-1/3$	-0.283	$+5.9 \times 10^{-14}$	subtracts	$4.9 \times 10^{28} (0.90)$
^{29}Si (enriched)	0	0.29	-0.555	-5.1×10^{-14}	adds	$5.0 \times 10^{28} (0.50)$
^{129}Xe (solid)	0	0.41	-0.778	-7.2×10^{-14}	adds	$1.8 \times 10^{28} (0.20)$
^3He (solid, 35 bar)	-0.056	0.87	-2.128	$+7.8 \times 10^{-14}$	subtracts	$2.5 \times 10^{28} (0.90)$

- Isotope bookkeeping.** A two-layer composite — a proton layer of thickness t_p polarized to $P_p = 0.90$, backed by a negative-moment isotope layer at areal-moment balance $f = 0.99$ — has $\sum_i n_i \mu_i P_i \simeq 0$ while retaining a substantial exotic charge. For $t_p = 3$ mm of polyethylene the balance requires 48 mm of Na^{15}N_3 azide or 12.6 mm of solid ^3He at 35 bar; the composite keeps $Q/Q_{\text{PE}} \simeq 0.50$ of the saturated exotic charge per unit area — the isotope term contributing $\leq 1\%$ of either sign at this t_p . With the corrected finite-geometry force (Section 3.3) the composite torque is $\tau_{\text{exo}} \simeq 2.9 \times 10^{-22}$ N·m in the maximum-reach torsion geometry — a factor ~ 180 *below* the 30-day thermal reach of that readout, and $\times 4$ below the pure-PE benchmark torque. Thinner proton layers dilute the signal further ($t_p = 1$ mm gives 5×10^{-23} N·m): the proton-layer thickness, not the isotope charge, is the cost driver. The composite’s role is therefore not to rescue the torsion signal-to-noise but to provide the phase detector of Section 5.2 with a source whose magnetization vanishes in-material. Inverting the negative-moment species is routine DNP targetry (microwave sideband choice) and is metastable at mK temperatures, with nuclear T_{1n} of hours to days; polarization inhomogeneity across the layer pair feeds directly into the material residual and is part of what the in-situ servo below must absorb. An electron-moment compensation variant (paramagnetic dopant cancelling the nuclear magnetization) was evaluated and rejected: the electron bath adds spin noise at f_0 , and the electron exotic coupling is not fixed by the benchmark, which would import model dependence into the source.
- In-material floor.** The far-field moment balance is exact, but the two moment layers sit at different depths, so the near fringe does not cancel identically. The exact charged-sheet magnetostatics of the layered composite gives a minimum residual $\delta_{\text{min}} \simeq 4 \times 10^{-2}$ of the uncompensated modulated torque, essentially flat in the scalar balance f around $f^* = 0.98$: a single scalar cannot null all multipoles of the depth-separated profile. δ_{min} is the purely material floor; an in-situ servo (SQUID pickup of the modulated fringe plus RF trim of the isotope polarization, read out independently of the torsion channel) contributes a further factor δ_{servo} .
- Superconducting interlayer.** A Meissner membrane in the $371 \mu\text{m}$ gap (source face at $z = 0$, shield at $a = 150 \mu\text{m}$, detector face at standoff $b = 170 \mu\text{m}$, disc radius $R = 0.30$ m)

blocks the magnetic fringe while the exotic force passes: the membrane is unpolarized and carries no spin content. The pendulum hangs *above* the shield, so the suspension never penetrates it — no aperture leakage, the usual weakness of shielded torsion geometry. The 2D analogue is solved exactly by the Joukowski conformal map with the streamline image system, verified to machine precision against the boundary condition and to four digits against the closed-form asymptote $S = 2R^2/((a+b)b)$: a localized source is attenuated by $S_{2D} = 3.3 \times 10^6$, while the plate charge-sheet (dipole) source — the physical model of the polarized plate — gives $S_{2D} \simeq 1.1 \times 10^{10}$; the 3D disc adds roughly one power of R/a ($S_{3D} \sim S_{2D}^{3/2}$, i.e. $\sim 6 \times 10^9$ – 10^{15}). In the torsion readout the membrane doubles as the grounded electrostatic guard of Section 5.4; in the phase-detector readout it is recomputed at the millimetre standoffs of that geometry (Section 5.2). Real-shield caveats — flux-pinning remanence in the 10 nT class, seams, and 3D edge leakage, with the plate fringe ring sitting only ~ 5 cm inside the shield rim — set the practical floor below this ideal-Meissner ceiling and must be measured in situ. Two qualifications are load-bearing. First, the quoted S_{3D} range spans six orders of magnitude: the 3D edge problem is *not* solved on paper, and these numbers are the ideal-boundary-condition ceiling, not a demonstrated shield factor. Second, the membrane must operate in the polarization holding field: at 0.3–0.5 T normal to its plane any type-II candidate (Nb: $H_{c1} \simeq 180$ mT) is deep in the vortex state, where pinning remanence and history-dependent fields replace the Meissner condition. Options — a reduced holding field below H_{c1} if the frozen-spin T_{1n} permits, or the in-plane-polarization variant of Section 3.3 (holding field parallel to the membrane plane, where thin-film vortex entry is suppressed, at the price of a factor of a few in signal) — are open. The pre-Phase-1 requirement list is: membrane material, thickness, H_{c1}/H_{c2} /critical current and operating temperature; a zero-field-cooled protocol; vortex penetration and pinning measured at the 0.1 Hz modulation frequency; a full 3D magnetostatic (finite-element) model of the finite disc including seams, mounts and the edge; an in-situ $S(f_0)$ demonstration in the actual holding field; and an analysis of Meissner-force loading, membrane vibration and the mechanical coupling to the modulated source.

- **Composite floor and required performance.** With $\delta_{\text{net}} = \delta_{\text{material}} \times \delta_{\text{servo}}$, the calibrated magnetic floor is $\text{CAL} \times \delta_{\text{net}} \times 4.5 \times 10^{-6}$ N·m/S, where $\text{CAL} = 10^{-3}$ is the polarimetry knowledge. With the corrected benchmark torque coverage of the pure-PE signal requires $S \geq 3.9 \times 10^{10}$ at $\delta_{\text{net}} = 10^{-2}$, rising to $S \geq 1.6 \times 10^{11}$ at the material floor $\delta_{\text{net}} = 4 \times 10^{-2}$ (thin composite: 9×10^{11} – 4×10^{12}). These requirements *exceed the ideal 2D dipole ceiling* $S_{2D} \simeq 1.1 \times 10^{10}$ quoted above — and the corrected benchmark torque in any case sits below the 30-day thermal reach of the torsion readout — so for the mechanical instrument this route no longer closes even on paper. Its realistic content is the exclusion program of Section 5.4; the benchmark-relevant incarnation of the same source is the phase detector of Section 5.2, whose suppression budget is computed there. Phase 1 must demonstrate δ_{net} and $S(f_0)$ in situ before any benchmark claim is made in either readout.

4 Bounds status, honestly split

Table 3 quotes each bound against the coupling it actually constrains. Margins must not be mixed: the SN 1987A limit applies to the quadratic combination $g_p^2 + 1.64 g_n^2 + 0.87 g_n g_p$, not to single couplings.

The SN 1987A bound would bite only at $c_\nu < 276$ and the HESS bound at $c_\nu < 1594$, so under the quoted bounds and conventions the entire scenario band $c_\nu \in [1/3, 3]$ lies below every line of Table 3. The pion tail is dead at millimetre separations, leaving the window clean for a tabletop test.

Table 3: Astrophysical constraints on the benchmark, with split margins (compilation and combination as in [4]; dense-matter uncertainties apply).

Channel	Predicted	Bound	Margin
SN 1987A (quadratic combo)	1.77×10^{-23}	1.35×10^{-18}	$\times 7.6 \times 10^4$
HESS J1731–347 (neutron star)	$ g_n = 1.76 \times 10^{-13}$	2.8×10^{-10}	$\times 1.6 \times 10^3$
WD cooling (g_{ee} , KSVZ loop)	4.8×10^{-20}	2.8×10^{-13}	$\times 5.8 \times 10^6$
Proton single-coupling (representative)	4.1×10^{-12}	$\sim 1.5 \times 10^{-9}$	$\times 3.6 \times 10^2$

5 The shielded nuclear-spin phase detector

Both readouts below share one hard component — the polarized source of Section 3.5 — and one hard problem — the magnetic fringe of that source (Section 3.4). They differ in what the exotic interaction is converted into: a precession-phase shift of a small nuclear-spin ensemble behind the superconducting membrane (the primary readout, Section 5.2) or a mechanical torque on a suspended plate (the alternative, Section 5.4). The phase readout removes the torsion fiber, the mechanical Q , the 0.1 Hz mechanical resonance, the large detector plate, the 371 μm mass drive and the patch-force/gravity-gradient coupling of the detector, while losing nothing load-bearing: with the corrected force of Section 3.3 the mechanical channel never reached the benchmark even at its thermal floor. What does *not* change is the magnetic problem — the membrane, the source compensation and the in-situ metrology remain the core of the program in either readout. The quantitative audit below is equally negative for the phase readout on the benchmark point (Section 5.7): the architecture is presented as the correct simplification and as a metrology/exclusion instrument, not as a benchmark detector. All numbers in this section are from `generator_project/calc_spin_phase_detector.py` (phase detector) and the scripts of Sections 3.3–3.5.

5.1 The polarized source

The source is common to both readouts: a thin (thickness $\sim 1\text{--}2$ cm) large-area plate of a proton-rich dielectric (benchmark: polyethylene), polarized to $\text{pol}_{\text{src}} \simeq 99.9\%$ (ideal proton thermal equilibrium); the practical route is dynamic nuclear polarization at 5 T/1 K with frozen-spin holding at 0.3–0.5 T and $\lesssim 100$ mK, with 20 T/5 mK nuclear demagnetization as the high-polarization fallback (see below); $\geq 90\%$ is retained as the conservative engineering target. The polarization must be *measured* to $\lesssim 0.1\%$, because the magnetic-calibrated floor of Section 3.4 is polarimetry-limited. Per Section 3.3 the signal is set by the facing area, so bulk mass beyond $\sim 5\lambda_\theta$ of depth is not purchased. The source carries the SQUID pickup loop and the RF trim of the isotope-layer polarization (the servo of Section 3.5); its task is the largest pseudoscalar spin density at the smallest external magnetic multipole moment.

Nuclear polarization is thermal: $\text{pol} = \tanh(\mu B/k_B T)$, where μ is the magnetic moment of the active nucleus. The benchmark material polarizes on its protons, $\mu = \mu_p = 2.793 \mu_N$; Table 4 quotes the conservative $\mu = \mu_N$ values, so the proton rows sit above them (at 20 T/5 mK, $x = 4.1$ and $\text{pol} = 99.9\%$). Achieving $\geq 90\%$ is within established nuclear-demagnetization practice.

The simpler baseline is dynamic nuclear polarization: in proton-rich plastics at 5 T/1 K it routinely delivers proton polarizations $\geq 90\%$ (established polarized-target practice [8]), and at the frozen-spin holding point the nuclear T_{1n} of hours to days makes the polarization quasi-static over a measurement cycle; the RF inversion of the isotope layer of Section 3.5 is performed at the same holding field. This relaxes the simultaneous 20 T/5 mK requirement of the polarization stage; Table 4 remains the thermal-equilibrium reference for the high- P fallback.

Table 4: Achievable nuclear polarization $\text{pol} = \tanh(\mu_N B/k_B T)$ quoted for one nuclear magneton (conservative lower bound; multiply x by μ/μ_N for the active nucleus — for the benchmark proton, $\mu_p/\mu_N = 2.793$).

B , T	T	x	pol
10	1 K	0.004	0.4%
10	10 mK	0.366	35%
20	5 mK	1.46	90%
30	1 mK	11.0	$\approx 100\%$

5.2 Readout principle and detector scenarios

Behind the membrane the exotic interaction enters the detector spin Hamiltonian as a pseudo-magnetic field, $H_{\text{eff}} = -\boldsymbol{\mu}_{\text{det}} \cdot \mathbf{B}_{\text{pseudo}}$: not an electromagnetic field, but the convenient unit for a spin-dependent energy shift. The readout is the CASPER-class spin-precession measurement [6] transplanted from the oscillating real field of dark matter to the static virtual exchange of a nearby polarized source. For the planar source the per-spin pseudo-energy at distance z behind the source face is $U(z) = n_s P \int_z^{z+t_s} W(u) du$ with the kernel W of Section 3.3; the level splitting of a fully polarized detector species A is $\Delta\omega_A = 2U(g_A/g_{\theta pp})/\hbar$ with the signed effective charges of Table 2. The reference geometry is a pancake cell of 5 cm diameter and 5 mm height (9.8 cm³) whose front face sits 2.5 mm behind the source face (source–membrane gap $a = 1.0$ mm, membrane–cell standoff $b = 1.5$ mm, the membrane doubling as the end-cap of the cell’s static field pocket below). For the 3 mm proton layer at $P = 0.90$: $U_{\text{avg}} = 2.3 \times 10^{-53}$ J, i.e. a proton-equivalent splitting $\Delta\omega_p = 4.3 \times 10^{-19}$ rad/s; for ^3He $\Delta\omega = -8.2 \times 10^{-21}$ rad/s, or $B_{\text{exo}} = 4.0 \times 10^{-29}$ T. The spin-projection floor of an ensemble of N spins polarized at P_{det} with coherence time T_2 after integration T is $\delta\omega = 1/(P_{\text{det}}\sqrt{NT_2T})$. Table 5 collects the scenarios.

Table 5: Spin-phase detector scenarios for the reference cell (9.8 cm³ at the 2.5 mm standoff, 30-day integration): exotic precession shift $\Delta\omega_{\text{exo}}$ and the resulting SNR against the spin-projection floor $\delta\omega = 1/(P_{\text{det}}\sqrt{NT_2T})$. Gas densities at 10 bar, 300 K; the solid ^3He row is speculative (exchange-narrowed $T_2 \sim 1$ s assumed).

Scenario	$g_A/g_{\theta pp}$	N	T_2 , s	$\Delta\omega_{\text{exo}}$, rad/s	SNR (30 d)
^3He gas, 10 bar (SEOP [10])	−0.019	2.4×10^{21}	10^4	-8.2×10^{-21}	3.8×10^{-5}
^{129}Xe gas, 10 bar	+0.017	2.4×10^{21}	10^3	$+7.5 \times 10^{-21}$	5.6×10^{-6}
^1H liquid (PHIP/d-DNP)	+1	6.6×10^{23}	3	$+4.3 \times 10^{-19}$	2.9×10^{-4}
^3He solid, 35 bar (option)	−0.019	2.5×10^{23}	1	-8.2×10^{-21}	5.9×10^{-6}

Two comagnetometer channels are available. In the ^3He – ^{129}Xe difference $\omega_{\text{He}}/\gamma_{\text{He}} - \omega_{\text{Xe}}/\gamma_{\text{Xe}}$ real magnetic fields cancel while the exotic shifts *add* — the effective charges of the two species carry opposite signs (Table 2) — giving $B_{\text{exo}}^{\text{diff}} = 1.4 \times 10^{-28}$ T against a combined floor of 1.8×10^{-23} T; a K– ^3He pair with a co-located alkali magnetometer reads the same B_{exo} against 1.0×10^{-24} T [11, 12]. The species-ratio rejection R_{cg} of real fields is an intrinsic stage independent of the membrane; we budget $R_{\text{cg}} = 10^3$ – 10^6 and require its in-situ demonstration at the modulation frequency.

Honesty requires the following statement: at the benchmark the quantum SNR of the best cell is $\sim 4 \times 10^{-5}$ in 30 days ($\times 3.5$ per year of integration), and the millimetre standoff penalizes short ranges — across the band the ^3He shift varies only from -6.6×10^{-21} rad/s ($c_\nu = 1/3$) to -2.3×10^{-21} rad/s ($c_\nu = 3$), because the c_ν^{-2} coupling gain is eaten by the $e^{-z/\lambda}$ penetration loss. The phase readout is therefore not a raw-sensitivity upgrade over the torsion thermal reach; it is a *systematics* upgrade — no mechanical channel at all, and an intrinsic rejection of real

magnetic fields — whose value is that every suppression factor demonstrated in situ converts one-to-one into exclusion reach (Section 5.3).

5.3 Modulation, magnetic budget and the static pocket

The detector is stationary; the modulation acts on the *source* and the lock-in runs at $f_0 \sim 0.01$ – 10 Hz. Four modes are computed (exact charged-sheet magnetostatics of the composite source at the cell centre, $z = 5.0$ mm; unshielded fringe 2.3×10^{-5} T for pure PE, 1.4×10^{-5} T for the $f = 0.99$ composite):

- **$\pm P$ global reversal (baseline):** both layers are RF-flipped together, so the magnetic balance is kept and only the residual swings sign, $B_{\text{mod}} = 1.4 \times 10^{-5}$ T, while the exotic charge flips fully. Required total suppression $\times 3.4 \times 10^{23}$.
- **Source z -scan (range shape):** a 0.5 mm displacement modulates the exotic energy by 10.5% but the magnetic fringe by only 1.7×10^{-4} relative — an intrinsic discrimination of $\times 600$ before any shielding, because the Yukawa profile and the power-law fringe have different z -dependence. Required suppression $\times 5.7 \times 10^{20}$, referenced to the modulated exotic fraction.
- **Lateral translation (± 0.5 mm):** at the centred cell B_z responds only at second order (3.4×10^{-11} T; the transverse B_x appears at first order, 1.2×10^{-9} T), while the exotic profile is translation-invariant over the cell (the rim is $\sim 60 \lambda_\theta$ away). This mode modulates only the magnetic background: it is the in-situ calibration channel.
- **RF inversion of the isotope layer alone:** swings the full imbalance ($B_{\text{mod}} = 6.8 \times 10^{-6}$ T) against $\leq 1\%$ of the exotic charge — rejected as a signal modulator, retained as the compensation diagnostic of Section 3.5.

The suppression chain has three measured factors: the membrane $S(f_0)$ — whose ideal 2D ceiling at this geometry is $S_{\text{mono}} = 4.8 \times 10^4$ for a localized source and $S_{\text{dip}} = 1.1 \times 10^8$ for the plate dipole (the 3D edge problem of Section 3.5 applies) — the servo depth δ_{servo} , and the comagnetometer rejection R_{cg} . With δ_{servo} a residual fraction the chain total is $X = S \times R_{\text{cg}} / \delta_{\text{servo}}$; the most optimistic combination, $S_{\text{dip}} \times 10^6 / 10^{-2} \simeq 1.2 \times 10^{16}$, still falls short of the benchmark leakage by $\times 3 \times 10^7$ in the baseline reversal mode and $\times 5 \times 10^4$ in the z -scan mode (with the localized-source ceiling and $R_{\text{cg}} = 10^3$: $\times 7 \times 10^{13}$ and $\times 10^{11}$). Direct benchmark coverage therefore requires suppression physics beyond the ideal membrane-plus-comagnetometer combination: the obstruction is not a pending engineering stage but the absence of a known physical mechanism closing the gap (Section 5.7). The realistic program is staged: the membrane is a *measured* component, $S(f) = B_{\text{no shield}} / B_{\text{with shield}}$, with remanence and vortex noise characterized before any exotic search; and every demonstrated factor converts directly into reach, since a null at total demonstrated suppression X excludes coupling products $\gtrsim 3.4 \times 10^{23} / X \times$ the benchmark ($\sim 3 \times 10^7 \times$ at the optimistic combination above; the quantum floor permits it, the coherent flux-noise requirement at f_0 being 10^{-17} – 10^{-15} T/ $\sqrt{\text{Hz}}$ depending on R_{cg} , against the 10^{-13} – 10^{-15} T/ $\sqrt{\text{Hz}}$ practice of Nb-film trapped-flux shields).

One requirement is new in this readout: the detector needs its own weak holding field (~ 10 μT) centimetres from the source's 0.3–0.5 T solenoid — a static attenuation of $\times 4 \times 10^4$ (quoted $\times 10^5$ with margin). This is a zero-field-cooled closed superconducting capsule (flux conservation, preceded in SQUID-NMR and low-field practice) whose end-cap is the membrane itself; SQUID readout is preferred, optical access through the capsule wall being a leak. The cell's back-action on the source is negligible ($\sim 10^{-5}$ A·m², static).

5.4 Alternative readout: the torsion balance

The alternative keeps the detector mechanical: a second large polarized plate suspended as a cryogenic torsion pendulum in the Eöt-Wash/spin-pendulum class [5, 7], driven by the modulated source at the pendulum resonance and read at the lock-in output. This is what the corrected force of Section 3.3 leaves of the original proposal: with the benchmark torque now below the thermal reach of the best configuration, the mechanical readout is an exclusion instrument and the commissioning apparatus of the magnetic program, not a benchmark test. Geometry-corrected signals (Section 3.3), resonant gain $Q = 10^5$ used consistently, 1 cm torque arm, $\text{pol} = 0.999$ (ideal proton thermal equilibrium at 20 T/5 mK; the conservative $\text{pol} = 0.90$ reduces every torque below by 19%):

- $A = 1000 \text{ cm}^2$, gap $d = \lambda_\theta$: $\tau_{\text{sig}} \simeq 5.3 \times 10^{-22} \text{ N}\cdot\text{m}$ — *below* the $10^{-18} \text{ N}\cdot\text{m}$ spin-torsion sensitivity by a factor $\sim 2 \times 10^3$; this configuration is a null-exclusion instrument (a null at the magnetic-calibrated floor of Section 3.4 excludes dipole–dipole coupling products $\gtrsim 1 \times 10^{13} \times$ the benchmark).
- $A = 1000 \text{ cm}^2$, gap $d = 0.1\lambda_\theta = 371 \text{ }\mu\text{m}$: $\tau_{\text{sig}} \simeq 1.15 \times 10^{-21} \text{ N}\cdot\text{m}$ — the maximum-reach configuration, at the price of sub-millimetre gap control between cryogenic polarized plates. Its nominal 30-day thermal SNR is 0.022 (the worst-case $\times 10$ tensor-structure uncertainty of Section 3.3 reduces it to 0.002): even at the thermal floor the benchmark is invisible to this readout, which with today’s magnetic audit delivers a calibrated null reach (Section 3.4).
- $A = 100 \text{ cm}^2$ (low-cost), gap $371 \text{ }\mu\text{m}$: $\tau_{\text{sig}} \simeq 1.15 \times 10^{-22} \text{ N}\cdot\text{m}$ — a commissioning/exclusion instrument.

For a pendulum with $I = 10^{-5} \text{ kg}\cdot\text{m}^2$, $f_0 = 0.1 \text{ Hz}$, $Q = 10^5$ at 4 K, the thermal torque noise floor is $S_\tau = \sqrt{4k_B T \kappa / (Q \omega_0)} \simeq 1.2 \times 10^{-16} \text{ N}\cdot\text{m}/\sqrt{\text{Hz}}$, i.e. a 30-day reach of $5.2 \times 10^{-20} \text{ N}\cdot\text{m}$: the thermal-limited SNR = 5 integration time for the maximum-reach signal is $\sim 1.4 \times 10^{11} \text{ s}$, so the thermal floor is not a practical operating point for a benchmark test. The realistic floor is set by systematics, dominated by:

- the ordinary magnetic torque of the polarized plates themselves (Section 3.4): $M = n_p \mu_p P \simeq 1.1 \times 10^3 \text{ A/m}$ ($\mu_0 M \simeq 1.4 \text{ mT}$) gives a modulated torsion-channel torque $\simeq 8.5 \times 10^{-4} \text{ N}\cdot\text{m}$ (with the resonant gain), $\sim 10^{18}$ above the exotic signal; it is flat across the gap scan, and its calibrated residual — polarimetry-limited at $\sim 5 \times 10^{-9} \text{ N}\cdot\text{m}$ — sets the real floor of this apparatus [7];
- magnetic leakage from the polarizing and holding magnets and residual dipole coupling to field gradients (compensation coils, the superconducting interlayer of Section 3.5, and a depolarized-source null cycle); eddy currents in shields and suspension;
- electrostatic patch forces at sub-millimetre gaps (guarded electrodes, contact-potential mapping);
- gravity-gradient coupling of the moving source mass (measured by unpolarized dummy runs at identical motion);
- seismic and vibrational pickup (standard low-frequency isolation).

The thermal numbers above are the ideal limit; the realistic floor is the magnetic-calibrated value of Section 3.4, $\sim 5 \times 10^{-9} \text{ N}\cdot\text{m}$ (polarimetry-limited), while $10^{-19} \text{ N}\cdot\text{m}$ remains an aspirational target conditional on the compensation routes of Section 3.5. Depolarizing the source suppresses both the exotic signal and the polarization-linked magnetic background; it leaves gravitational, electrostatic, vibrational, and external-field systematics available for control measurements. It therefore tests spin dependence, not the exotic origin of a polarization-linked signal. Polarization reversal reverses the signal sign.

5.5 Low-cost 100 g-scale variant

The area-based scaling of Section 3.3 changes the economics in a convenient way: since only the facing skin is active, the low-cost variant is simply a smaller plate with the same cryomagnetic stack. With resonant integration the minimum detectable torque drops as $\tau_{\min}(t) = S_\tau/\sqrt{2t}$, reaching 5.2×10^{-20} N·m over 30 days at the thermal floor; the realistic floor is the magnetic-calibrated $\sim 5 \times 10^{-9}$ N·m of Section 3.4. The geometry-corrected variants (all numbers from `generator_project/calc_slab_geometry.py`):

Table 6: Low-cost variants at $Q = 10^5$ lock-in, $\text{pol} = 0.999$ (ideal proton equilibrium; conservative $\text{pol} = 0.90$ reduces τ and SNR of the upper rows by 19%), 30-day integration. SNR is thermal. “Null excludes” quotes the coupling-product reach of a null result at the magnetic-calibrated floor 5×10^{-9} N·m of Section 3.4 relative to the benchmark. The reach is quoted for the product of the two couplings; for a single symmetric coupling $g_1 = g_2 = g$ take the square root (e.g., $\sim 2 \times 10^6 \times$ the benchmark g for the 1000 cm² row). The last row is the isotope composite of Section 3.5, quoted directly at its engineering polarization $P_p = P_A = 0.90$ (the 19% ideal-to-conservative reduction does not apply to it): its required shielding exceeds the ideal ceiling of Section 3.5, so it does not close on the benchmark in this readout either; the composite’s live role is as the phase-detector source (Section 5.2).

Variant	τ_{sig} , N·m	SNR (30 d)	vs floor	null excludes
100 cm ² , λ_θ gap	5.3×10^{-23}	1.0×10^{-3}	$\times 1.1 \times 10^{-14}$	$\gtrsim 9 \times 10^{13} \times$ benchmark
100 cm ² , 371 μm	1.15×10^{-22}	2.2×10^{-3}	$\times 2.3 \times 10^{-14}$	$\gtrsim 4 \times 10^{13} \times$ benchmark
1000 cm ² , 371 μm	1.15×10^{-21}	0.022	$\times 2.3 \times 10^{-13}$	$\gtrsim 4 \times 10^{12} \times$ benchmark
1000 cm ² composite, 371 μm	2.9×10^{-22}	5.5×10^{-3}	$\times 5.7 \times 10^{-14}$	benchmark not covered

The honest statement: at the magnetic-calibrated floor, nulls exclude coupling products $\gtrsim 4 \times 10^{12} - 9 \times 10^{13} \times$ the benchmark depending on configuration — a new direct measurement regime for millimetre-range spin-spin forces, but not coverage of the benchmark itself, which is not closed even on paper in this readout (the required shielding of Section 3.5 exceeds the ideal 2D ceiling). The maximum-reach configuration remains the 1000 cm² plate at 371 μm . The honest cost statement: the superconducting magnet dominates the cost in all variants; the low-cost line is a factor of a few cheaper, not two orders, and its advantage is that it fits existing cryomagnetic infrastructure.

5.6 Phases

The staging is detector-agnostic through Phase 2: everything up to the spin cell is the source-and-shield metrology on which either readout rests.

- **Phase 0** (existing infrastructure): verify the 3.71 mm window is pion-clean at the relevant nuclear densities; reproduce the base-case torque calculation (`calc_spin_fifth_force.py`), the finite-geometry kernel and its validation against the exact slab identity and a direct 3D pair sum (`calc_slab_geometry.py`), the order-of-magnitude magnetic wall (`calc_magnetic_background.py`) and the closed-form magnetostatics audit (`calc_magnetic_compensation.py`); reproduce the isotope-composite bookkeeping and layered-composite floor (`calc_spin_compensation.py`), the exact 2D shielding solution (`calc_shield_attenuation.py`) and the full phase-detector budget — pseudo-energies, detector scenarios, comagnetometer channels, modulation modes, suppression chain and static pocket (`calc_spin_phase_detector.py`).
- **Phase 1** (source and shield as measured components): the polarized composite source with its SQUID pickup loop and RF trim, separated by the superconducting membrane from

a magnetometry station — no spin detector yet. The goal is the measurement program itself: the real magnetic compensation of the composite source (residual modulation and in-situ depth δ_{net}), the real shield factor $S(f)$ across the 0.01–10 Hz lock-in band in the operating holding field, and the remanence, trapped flux and vortex/pinning noise at f_0 — the numbers on which any later claim of either readout rests.

- **Phase 2** (classical verification): an atomic/SQUID magnetometer at the cell position confirms that the ordinary source-induced magnetic modulation is suppressed to the demonstrated level, $B_{\text{mod}}/S(f_0)$, and that the lateral-drive calibration mode of Section 5.3 behaves as computed. The membrane is at this point a *measured* component, not an idealized one, and the measured factors already do physics work: they set direct limits on stronger millimetre-range spin–spin interactions and define the engineering constraint that any future source material with lower external magnetization must meet.
- **Phase 3** (the spin cell): install the hyperpolarized $^3\text{He}/^{129}\text{Xe}$ cell with its comagnetometer readout and static pocket (Sections 5.2–5.3); run the $\pm P$ reversal as the baseline lock-in, the source z -scan as the range-shape channel and the lateral drive as the continuous background calibration. A null at total demonstrated suppression X excludes coupling products $\gtrsim 3.4 \times 10^{23}/X \times$ the benchmark; at the optimistic component combination of Section 5.3 this is $\sim 3 \times 10^7 \times$, and every measured factor converts one-to-one into reach. In parallel or instead, the torsion variant of the same source (Section 5.4) runs its exclusion program to $\sim 4 \times 10^{12} \times$ the benchmark coupling product ($\sim 2 \times 10^6 \times$ in a single symmetric coupling) at the maximum-reach geometry.
- **Phase 4 (conditional)**: benchmark coverage requires closing a suppression shortfall of $\times 5 \times 10^4$ – 3×10^7 in the most optimistic reading and $\times 10^{11}$ – 7×10^{13} in the strict one (Section 5.3) — a gap for which no known physical mechanism exists (Section 5.7). This is therefore not a deferred engineering stage but a research program on new suppression physics, whose candidate directions are the verdict list of Section 5.7; it becomes a quantitative proposal only after the Phase-1/2 measurements, together with a nuclear-spin calculation (or measurement) of the source’s effective charges beyond the decomposition-level accuracy of Table 2.

5.7 The no-go verdict and what could overturn it

The benchmark is inaccessible to the present source–detector family, and the obstruction is quantitative, not rhetorical: it is reached independently in three channels (Table 7). The quantum and thermal gaps are matters of scale that neither integration time nor detector size closes within this architecture. The magnetic gap is qualitatively different: the required suppression exceeds the most optimistic ideal-component product by $\times 5 \times 10^4$ – 3×10^7 (Section 5.3), and *no known physical mechanism closes the remaining gap* — the obstruction is not a pending engineering stage but an absent mechanism. This is the central result of the feasibility audit: the shielded nuclear-spin phase-readout architecture removes the mechanical limitations of torsion balances, but the benchmark remains inaccessible with the present source, spin-projection sensitivity and demonstrated magnetic-suppression physics.

What would overturn the verdict is not another incremental gain in geometry or integration time but one of the following qualitative changes:

- a **source** whose magnetic multipoles are suppressed by orders of magnitude below the isotope-composite floor of Section 3.5, while a macroscopic pseudoscalar nuclear-spin charge survives;
- a **detector** with a radically larger NT_2 than the compact gas cell of Table 5 — the quantum gap is $\times 7 \times 10^8$ in NT_2T , beyond what cell diameter or integration time can

Table 7: The three independent gaps of the no-go verdict (Section 5.7) at the θ -seesaw benchmark ($c_\nu = 1$), 30-day integration; entries from `calc_spin_phase_detector.py` and `calc_slab_geometry.py`. Each gap alone suffices.

Channel	Gap	What closing it requires
Quantum (phase readout)	$B_{\text{exo}} = 4.0 \times 10^{-29}$ T vs spin-projection floor 1.0×10^{-24} T: $\times 2.6 \times 10^4$ (SNR 4×10^{-5})	NT_2T up $\times 7 \times 10^8$ — time-only 6×10^7 yr, size-only cell diameter $\times 2.6 \times 10^4$
Thermal (torsion readout)	$\tau_{\text{exo}} = 1.1 \times 10^{-21}$ vs thermal reach 5.2×10^{-20} N·m: SNR 0.022	170 yr for thermal SNR = 1; the magnetic-calibrated floor is $\times 4 \times 10^{12}$ above the signal
Magnetic (either readout)	required $\times 3.4 \times 10^{23}$ ($\pm P$) / $\times 5.7 \times 10^{20}$ (z -scan) vs ideal chain 1.2×10^{16}	shortfall $\times 5 \times 10^4$ – 3×10^7 (strict: $\times 10^{11}$ – 7×10^{13}); no known mechanism

supply;

- a **new observable** separating the pseudoscalar channel from the magnetic one by a symmetry signature rather than by an amplitude difference behind a shield;
- a **channel pivot**: the companion photon/haloscope program [2], which then carries the benchmark-discovery burden if its sensitivity budget proves substantially better than the spin channel’s.

Until one of these materializes, the live content of the spin-side program is the metrology-and-exclusion program of Section 5.6: the measured compensation, shield factor and noise floor convert one-to-one into exclusion reach on stronger millimetre-range spin–spin interactions and into the engineering constraint for any future low-magnetization source material.

6 Two-lab consistency

The sharpest structural prediction is not the force magnitude but the *consistency condition*: the haloscope line at $f_\theta = 12.86$ GHz and the force range $\lambda_\theta = 3.71$ mm are the same mass $m_\theta = hf_\theta/c^2 = \hbar/(\lambda_\theta c)$. Measuring m_θ in either laboratory fixes the other laboratory’s target with zero free parameters; the cross-lab ratio $g_p/g_{\theta\gamma\gamma} = -198$ GeV is c_ν -independent. The force laboratory does not test a single gap: it measures the gap dependence $F(d)$ over a range spanning the sub-millimetre region and distances of order λ_θ , and the extracted range is compared with the mass measured by the haloscope. In the phase-detector readout the gap scan is the source z -scan of Section 5.3, whose exotic pseudo-field modulation of 10.5% per 0.5 mm at the reference standoff is a direct range measurement with an intrinsic $\times 600$ rejection of the magnetic fringe. A haloscope detection whose line implies $\lambda_\theta = 3.71$ mm while the range extracted from the gap scan disagrees (or vice versa) excludes the single-field embedding. With the no-go verdict of Section 5.7 the discovery burden for the benchmark itself passes to the haloscope; the force program retains this consistency test together with the exclusion reach of Section 5.6.

7 Falsifiability and predictions

1. **Range**: the force falls as e^{-r/λ_θ} with $\lambda_\theta = 3.71 c_\nu$ mm; a measurement of the range fixes c_ν and, through Table 1, all couplings.
2. **Spin dependence**: the force vanishes for unpolarized bodies and reverses sign under polarization reversal; any range-independent or spin-independent residual is not this effect.

3. **Null interpretation:** a null result at the magnetic-calibrated floor excludes dipole–dipole coupling products $\gtrsim 4 \times 10^{12} \times$ the benchmark in the maximum-reach torsion geometry (Section 5.4); in the phase-detector readout the exclusion is quoted per demonstrated suppression factor, $\gtrsim 3.4 \times 10^{23}/X \times$ the benchmark (Section 5.3). Exclusion of the benchmark itself is not achieved by either readout with ideal components and is the conditional content of Phase 4. All geometry numbers in this paper are computed at $c_\nu = 1$; across the band $c_\nu \in [1/3, 3]$ both the gap in Compton units and the coupling change ($F/A \propto c_\nu^{-4}$ at fixed physical gap), so the band-wide statement rests on the sensitivity map reproduced by `calc_slab_geometry.py` — at the fixed 371 μm gap the signal torque spans $7.7 \times 10^{-20} \text{ N}\cdot\text{m}$ ($c_\nu = 1/3$) down to $1.5 \times 10^{-23} \text{ N}\cdot\text{m}$ ($c_\nu = 3$) — not on a single point.
4. **Consistency:** the 12.86 GHz line and the 3.71 mm range must coexist; absence of either falsifies the single-field embedding.

Reproducibility

All numerical statements in this paper are produced by standard-library scripts in the repository [1]: `generator_project/calc_spin_fifth_force.py` (couplings, bounds, pointlike base-case force), `generator_project/calc_slab_geometry.py` (finite-geometry convolution, gap lever, grid-convergence check, c_ν sensitivity map, revised torques) and `generator_project/calc_stasis_mass_energy_gravdet.py` (polarization roadmap, noise budget, scaling tables); the classical magnetic background of Section 3.4 is audited by `generator_project/calc_magnetic_background.py` (order-of-magnitude wall) and `generator_project/calc_magnetic_compensation.py` (closed-form magnetostatics of the finite plates, gap-scan and sandwich compensation scans, polarimetry-limited floor); the candidate routes of Section 3.5 are computed by `generator_project/calc_spin_compensation.py` (isotope bookkeeping, layered-composite magnetostatics, in-material floor) and `generator_project/calc_shield_attenuation.py` (exact 2D conformal-map shielding factor, boundary-condition and closed-form cross-checks); the phase-detector program of Section 5 is computed by `generator_project/calc_spin_phase_detector.py` (per-spin pseudo-energy at the cell, detector scenarios, comagnetometer channels, the four modulation modes, the suppression chain, the static pocket and the no-go bookkeeping of Table 7), whose geometry integral is cross-validated against both the convolution and the exact slab identity of `calc_slab_geometry.py`. Rerunning them reproduces every quoted number.

References

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